

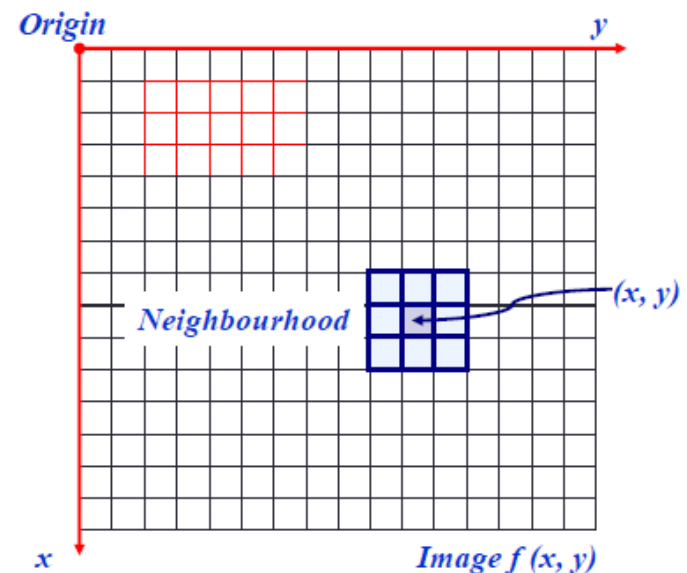


UNIT 2

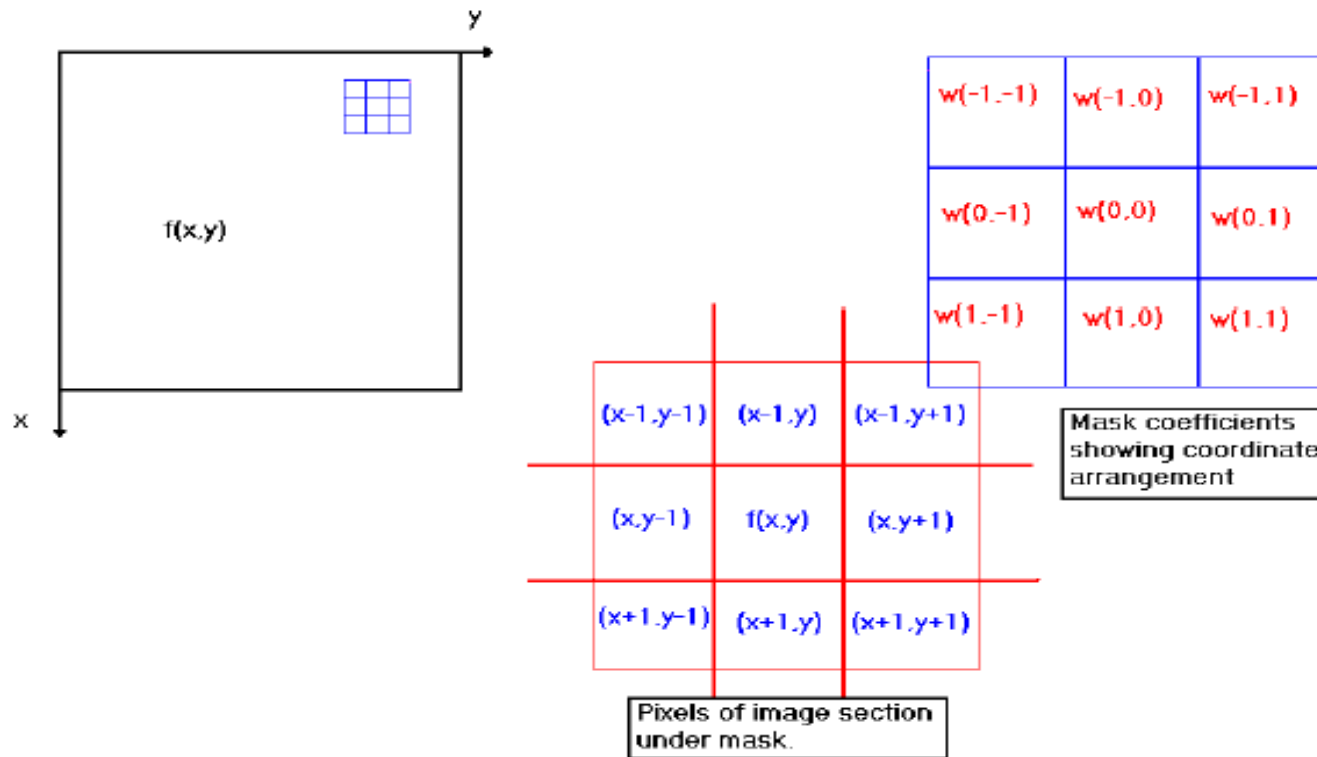


BASICS OF SPATIAL FILTERING

- Spatial masks or kernels or templates or windows are used
- Spatial Filtering operations are performed directly on the pixels of an image
- The process of spatial filtering consists simply of moving the filter mask from point to point in an image
-



The Mechanism of Spatial Filtering



For the 3X3 mask, the result (or response), R , of linear filtering with the filter mask at a point (x, y) in the image is

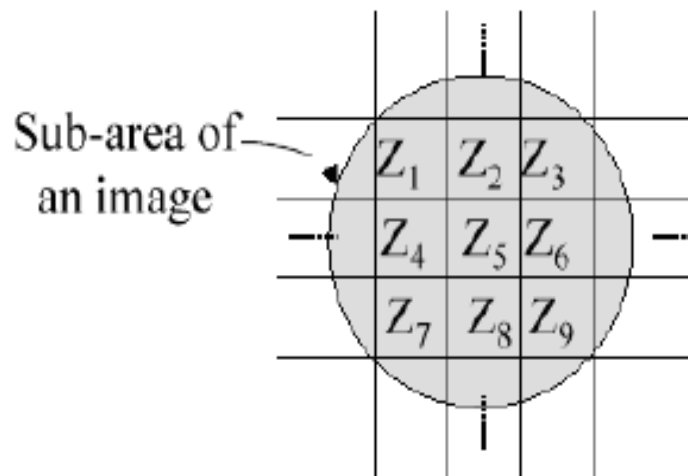
$$R = w(-1, -1)f(x - 1, y - 1) + w(-1, 0)f(x - 1, y) + \dots + w(0, 0)f(x, y) + p + w(1, 0)f(x + 1, y) + w(1, 1)f(x + 1, y + 1)$$



- A mask is a small (say 3x3) 2-D array, in which the values of the mask coefficients determine the nature of the process.

In general, one can assign different weight to each pixel and actually perform different operations on each pixel.

$$Z = w_1 Z_1 + w_2 Z_2 + \dots + w_9 Z_9 = \sum_{i=1}^9 w_i Z_i$$



w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

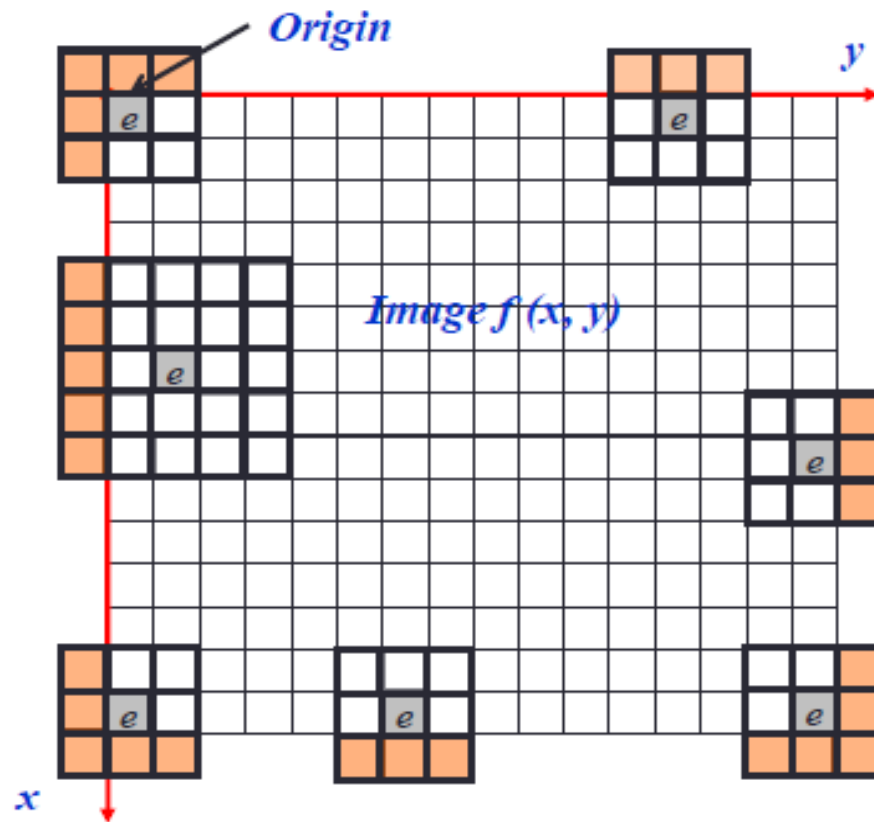
3 x 3 mask

A neighborhood



We Miss Pixels At The Edges!

However, at the edges of an image we miss pixels to form a neighborhood.



Smoothing Spatial Filters

- Smoothing filters are used for blurring and for noise reduction
 - Blurring is used in preprocessing steps
- Noise reduction can be accomplished by blurring with a linear filter and also by nonlinear filtering

Smoothing Linear Filters

- The output of a smoothing, linear spatial filter is simply the average of the pixels contained in the neighborhood of the filter mask
- Smoothing filters are called averaging filters
- They are also called lowpass filters



- Random noise typically consists of sharp transitions in gray levels
- Smoothing filters replace the value of every pixel in an image by the average of the gray levels in the neighborhood defined by the filter mask
- This process results in an image with reduced “sharp” transitions in gray levels
- ➔ Main application of smoothing filters is noise reduction
- However, edges also are characterized by sharp transitions in gray levels
 - Averaging filters have the undesirable side effect that they blur edges



- Another application of this type of smoothing includes the smoothing of false contours that result from using an insufficient number of gray levels
- A major use of averaging filters is in the reduction of “irrelevant” detail in an image
 - Irrelevant → pixel regions that are small with respect to the size of the filter mask



Two 3X3 smoothing (averaging) filter masks. The constant multiplier in front of each mask is equal to the sum of the values of its coefficients, as is required to compute an average

 $\frac{1}{9} \times$

1	1	1
1	1	1
1	1	1

Mask 1

 $\frac{1}{16} \times$

1	2	1
2	4	2
1	2	1

Mask 2



- An $m \times n$ mask would have a normalizing constant equal to $1/mn$
- A spatial averaging filter in which all coefficients are equal is sometimes called a box filter (mask1)
- Mask 2 yields a weighted average,
 - ➔ pixels are multiplied by different coefficients, thus giving more importance (weight) to some pixels at the expense of others.
- In mask 2, the pixel at the center of the mask is multiplied by a higher value than any other, thus giving this pixel more importance in the calculation of the average



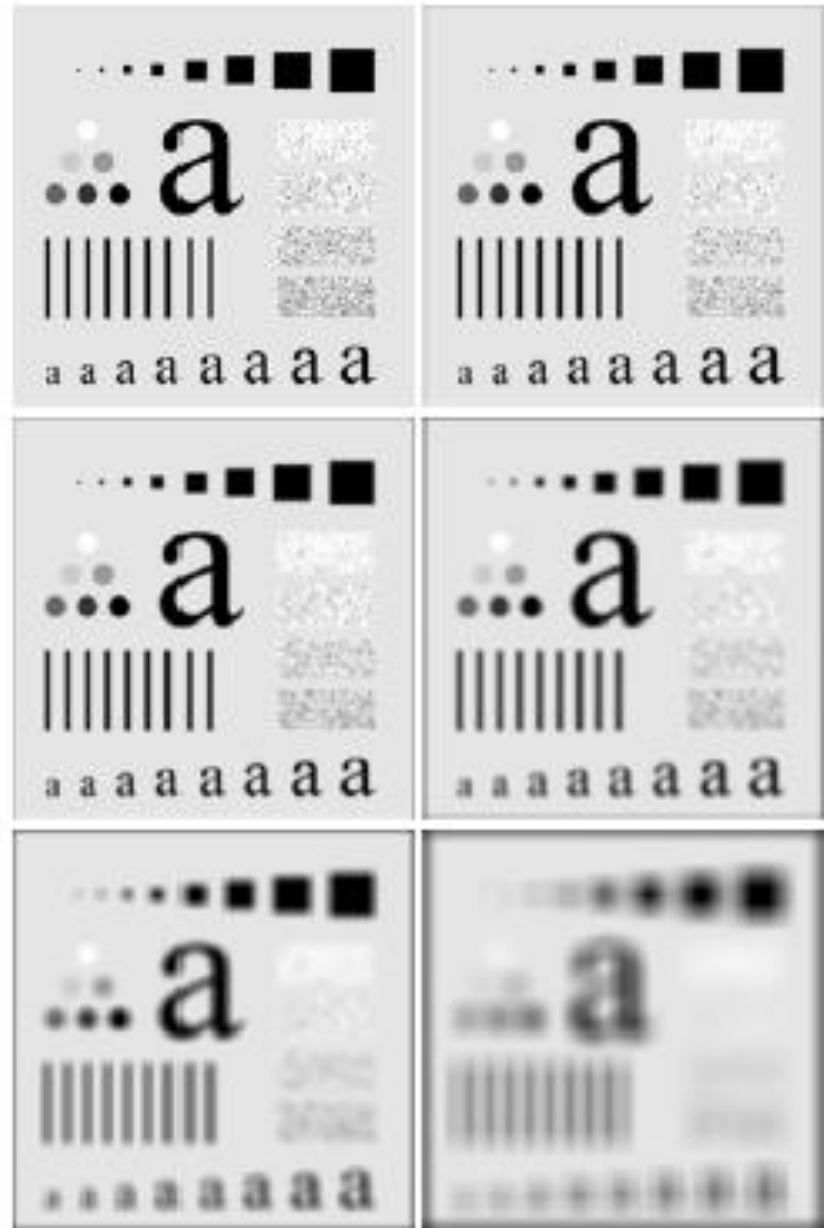
- General implementation for filtering an $M \times N$ image with a weighted averaging filter of size $m \times n$ (m and n odd) is given by the expression

$$g(x, y) = \frac{\sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)}{\sum_{s=-a}^a \sum_{t=-b}^b w(s, t)}$$

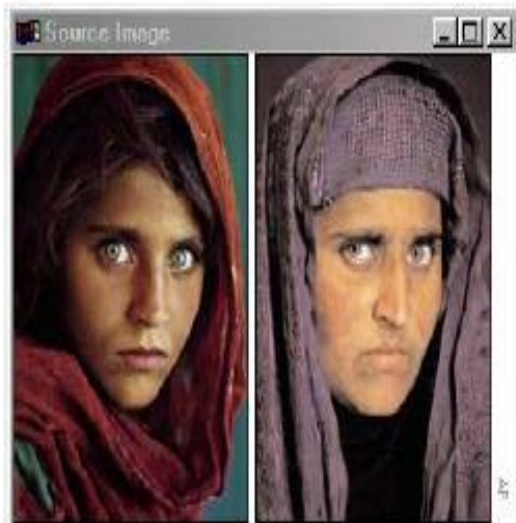


THE EFFECTS OF SMOOTHING AS A FUNCTION OF FILTER SIZE

Original image, of size 500x500 pixels.
(b)–(f) Results of smoothing
with square averaging filter masks of
sizes $n=3, 5, 9, 15$, and 35 ,
respectively



original



3x3



5x5



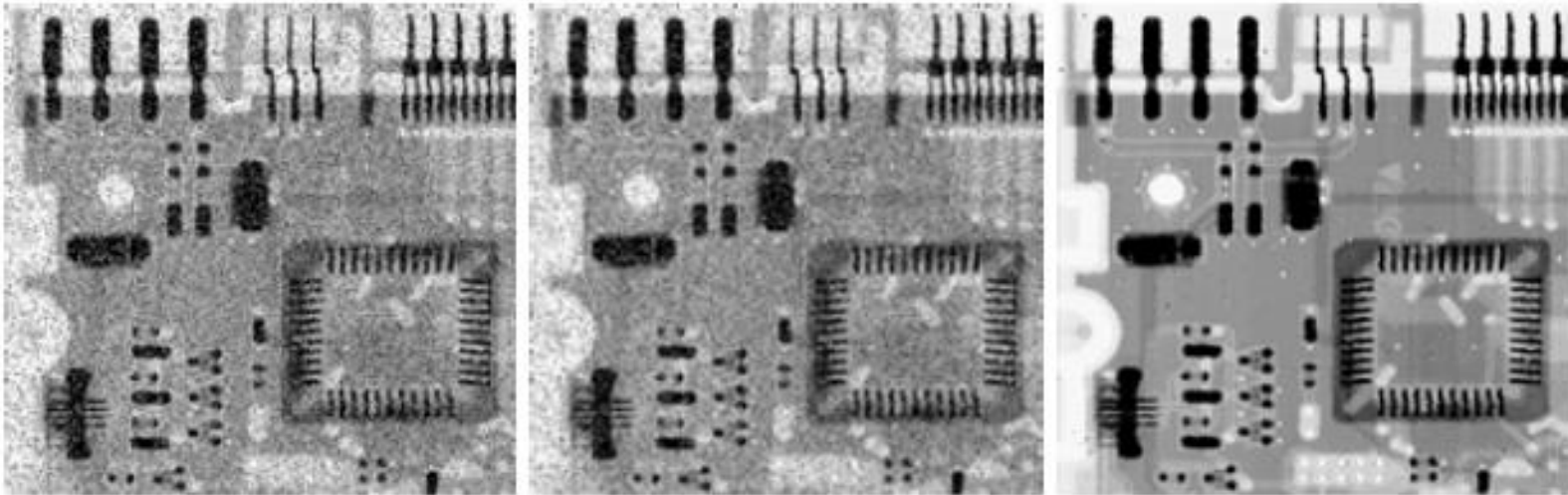
7x7



ORDER-STATISTICS FILTERS

- Order-statistics filters are nonlinear spatial filters
- Response is based on ordering (ranking) the pixels contained in the image area covered by the filter
- Value of the center pixel is replaced with the value determined by the ranking result
- The best-known example in this category is the *median filter*

- Median filters provide excellent noise-reduction capabilities, with considerably less blurring than linear smoothing filters of similar size
- Median filters are particularly effective in the presence of impulse noise (salt-and-pepper noise) because of its appearance as white and black dots superimposed on an image
- The median, ξ , of a set of values is such that half the values in the set are less than or equal to ξ , and half are greater than or equal to ξ



(a) X-ray image of circuit board corrupted by salt-and-pepper noise. (b) Noise reduction with a 3x3 averaging mask. (c) Noise reduction with a 3x3 median filter

SHARPENING SPATIAL FILTERS

- The principal objective of sharpening is to highlight fine detail in an image or to enhance detail that has been blurred
- Uses of image sharpening
 - Electronic printing
 - Medical imaging
 - Industrial inspection
 - Autonomous guidance in military systems
- Sharpening could be accomplished by spatial differentiation

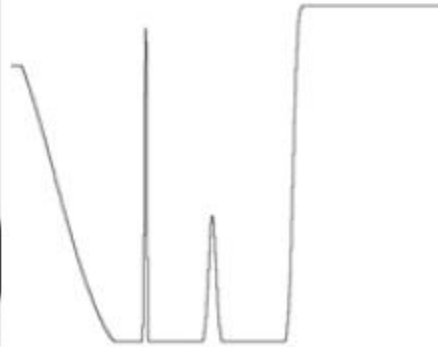
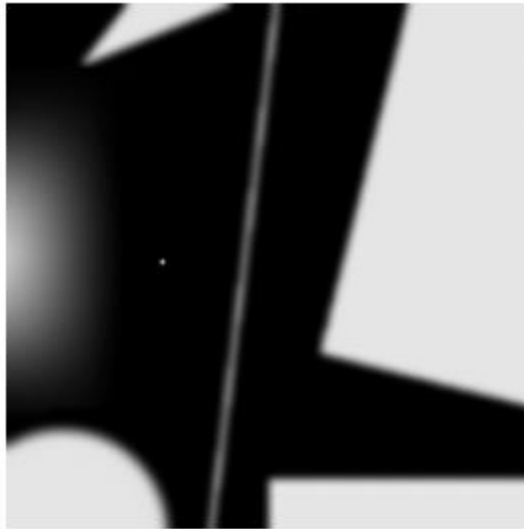
- The derivatives of a digital function are defined in terms of differences
- Definition for a first derivative
 - (1) must be zero in flat segments (areas of constant gray-level values);
 - (2) must be nonzero at the onset of a gray-level step or ramp; and
 - (3) must be nonzero along ramps
- Definition of a second derivative
 - (1) must be zero in flat areas;
 - (2) must be nonzero at the onset and end of a gray-level step or ramp; and
 - (3) must be zero along ramps of constant slope

- A basic definition of the first-order derivative of a one-dimensional function $f(x)$ is the difference

$$\frac{\partial f}{\partial x} = f(x + 1) - f(x)$$

- Similarly, a second-order derivative is defined as the difference

$$\frac{\partial^2 f}{\partial x^2} = f(x + 1) + f(x - 1) - 2f(x)$$



(a) A simple image (b) 1-D horizontal graylevel profile along the center of the image and including the isolated noise point. (c) Simplified profile (the points are joined by dashed lines to simplify interpretation).

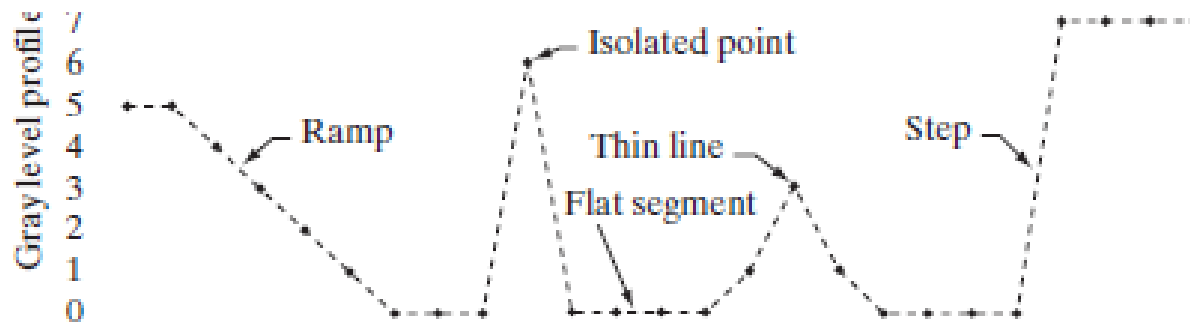
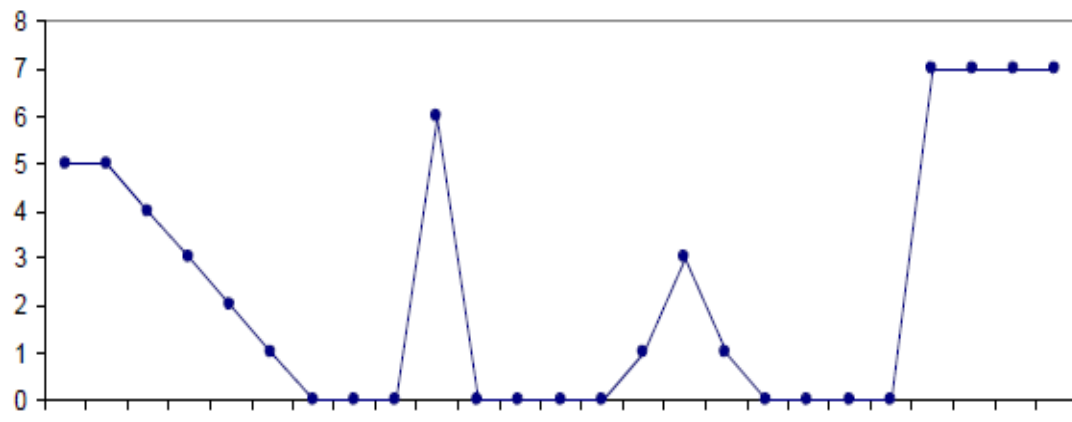


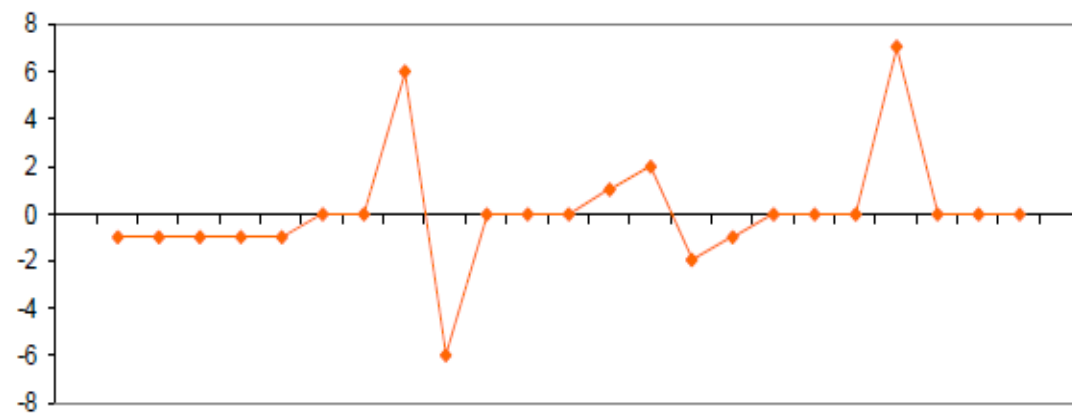
Image strip	5	5	4	3	2	1	0	0	0	6	0	0	0	0	1	3	1	0	0	0	0	7	7	7	7	*	*
First Derivative	-1	-1	-1	-1	-1	0	0	6	-6	0	0	0	1	2	-2	-1	0	0	0	7	0	0	0				
Second Derivative	-1	0	0	0	0	1	0	6	-12	6	0	0	1	1	-4	1	1	0	0	7	-7	0	0				

First Derivative



5 5 4 3 2 1 0 0 0 6 0 0 0 0 1 3 1 0 0 0 0 7 7 7 7

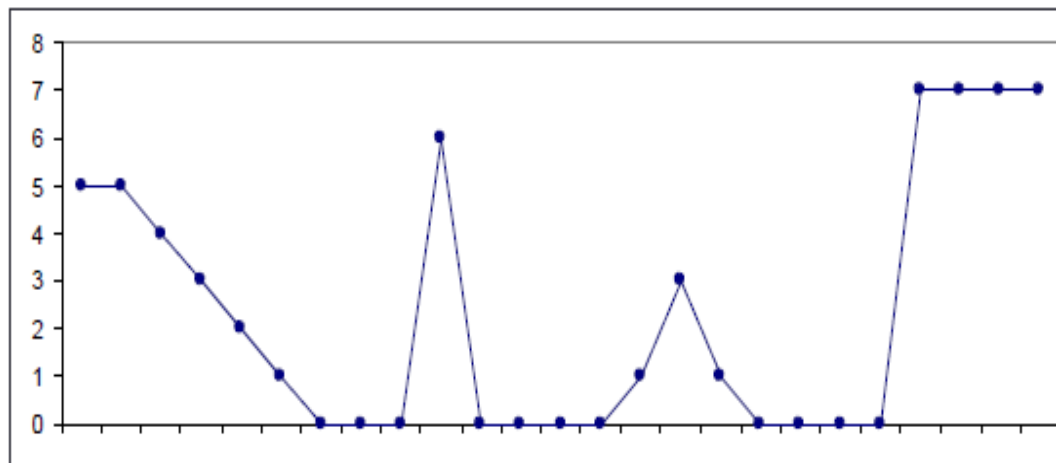
-1 -1 -1 -1 -1 0 0 6 -6 0 0 0 1 2 -2 -1 0 0 0 7 0 0 0



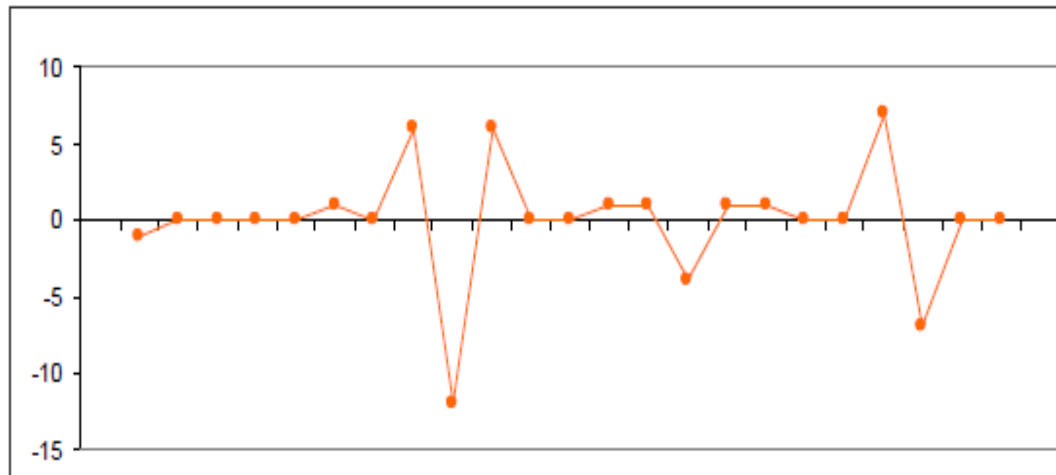
First derivative:

- 0 in constant gray segments
- Non-zero at the onset of steps or ramps
- Non-zero along ramps

Second Derivative



-1	0	0	0	0	1	0	6	-12	6	0	0	1	1	-4	1	1	0	0	7	-7	0	0
----	---	---	---	---	---	---	---	-----	---	---	---	---	---	----	---	---	---	---	---	----	---	---



Second derivative:

- 0 in constant gray segments
- Non-zero at the onset and end of steps or ramps
- 0 along ramps of constant slope.

- Comparing the response between first- and second-order derivatives, the following are the conclusions
 - (1) First-order derivatives generally produce thicker edges in an image
 - (2) Second-order derivatives have a stronger response to fine detail, such as thin lines and isolated points
 - (3) First order derivatives generally have a stronger response to a gray-level step
 - (4) Second- order derivatives produce a double response at step changes in gray level

USE OF SECOND DERIVATIVES FOR ENHANCEMENT— THE LAPLACIAN

- The approach basically consists of
 - Defining a discrete formulation of the second-order derivative
 - Constructing a filter mask based on that formulation
- The second derivative is more useful for image enhancement than the first derivative
 - Stronger response to fine detail
 - Simpler implementation
- The *Laplacian*, which is the first sharpening filter is
 - Isotropic : independent of direction of discontinuities in the image OR *rotation invaiant*
 - One of the simplest sharpening filters

- The *Laplacian* for a function of two variables $f(x,y)$ is defined as:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

- Consider the partial second-order derivative in the x -direction:

$$\frac{\partial^2 f}{\partial x^2} = [f(x+1, y) - f(x, y)] - [f(x, y) - f(x-1, y)]$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) - f(x, y) - f(x, y) + f(x-1, y)$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

- Similarly the second order derivative in the y -direction is given by:

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

- The two-dimensional Laplacian is obtained by summing these two components:

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x, y)$$

0	1	0
1	-4	1
0	1	0

1	1	1
1	-8	1
1	1	1

Positive Center Laplacian Mask

0	-1	0
-1	4	-1
0	-1	0

$$\nabla^2 f = 4z_5 - (z_2 + z_4 + z_6 + z_8)$$

-1	-1	-1
-1	8	-1
-1	-1	-1

$$\nabla^2 f = 8z_5 - (z_1 + z_2 + z_3 + z_4 + z_6 + z_7 + z_8 + z_9)$$

Problems with The Laplacian

- The Laplacian generally is not used in its original form for edge detection for several reasons:
 - As a second-order derivative, the Laplacian typically is unacceptably sensitive to noise.
 - The magnitude of the Laplacian produces double edges, an undesirable effect because it complicates segmentation.
 - Finally, the Laplacian is unable to detect edge direction

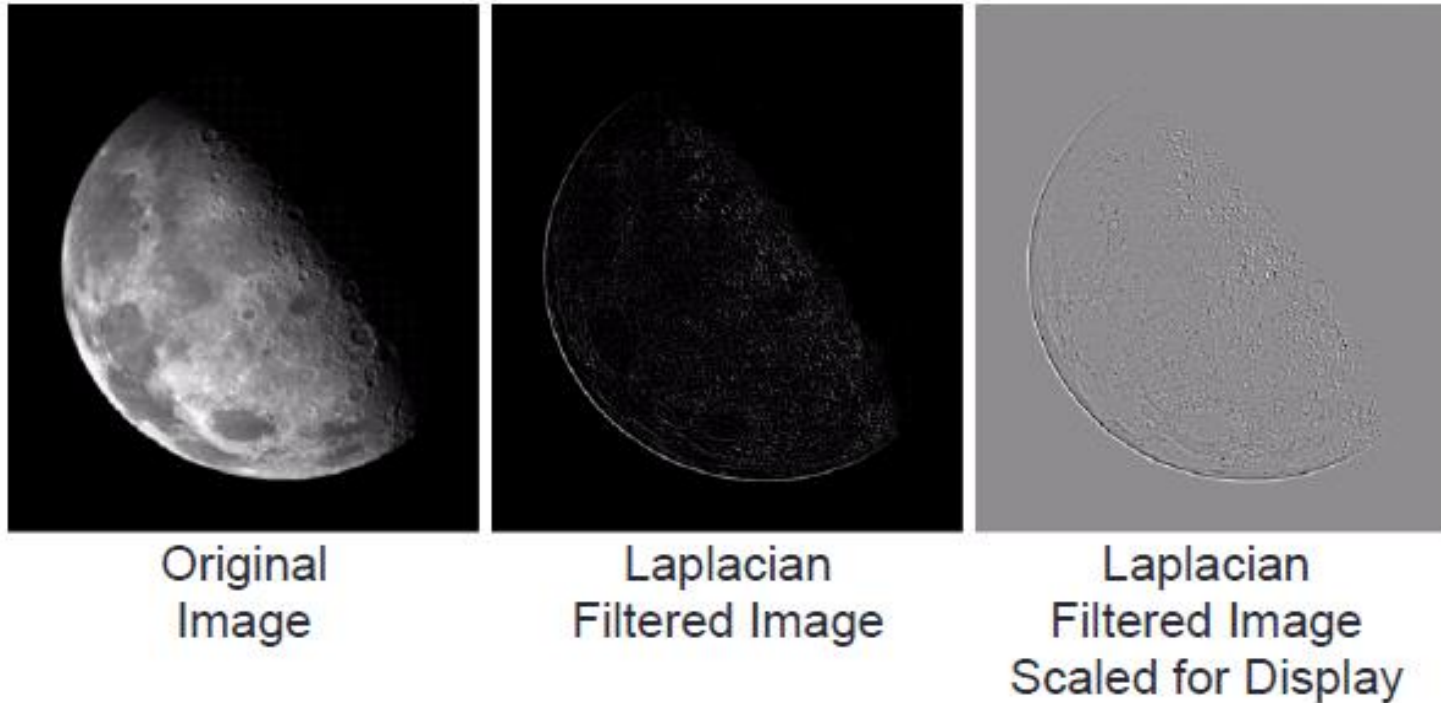


Image corrupted with
pepper noise



Output image after
applying Laplacian

- Applying the Laplacian to an image we get a new image that highlights edges and other discontinuities



- The result of a Laplacian filtering is not an enhanced image.

- Because the Laplacian is a derivative operator,
 - Highlights gray-level discontinuities in an image
 - Deemphasizes regions with slowly varying gray levels
 - Produce images that have grayish edge lines and other discontinuities, all superimposed on a dark, featureless background
- Background features can be “recovered” while still preserving the sharpening effect of the Laplacian operation simply by **adding the original and Laplacian images**
 - Negative center coefficient => subtract the Laplacian image to obtain a sharpened result.

- If the center coefficient of the Laplacian is *negative*, then

$$g(x, y) = f(x, y) - \nabla^2 f(x, y)$$

$$g(x, y) = f(x, y) - [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] + 4f(x, y)$$

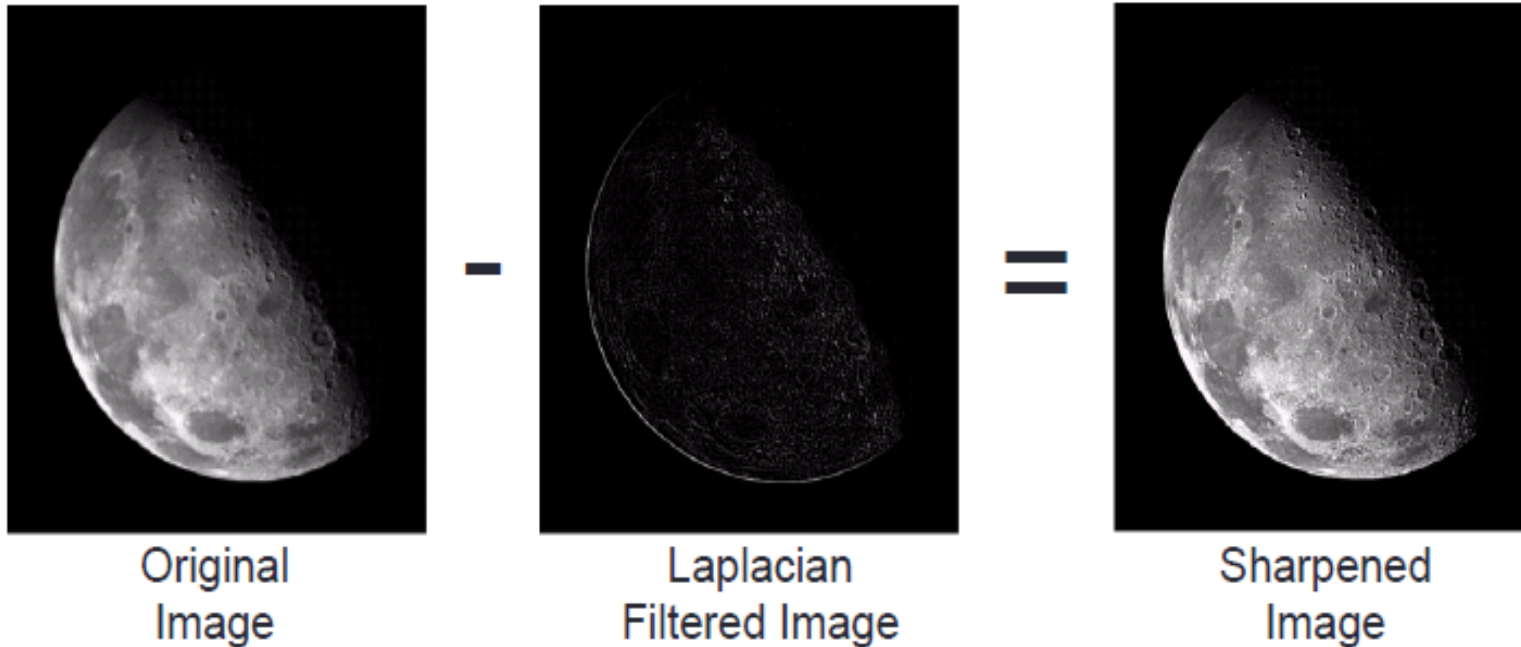
- This equation can be implemented using the followings masks

$$g(x, y) = 5f(x, y) - [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)]$$

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) & \text{If centre coefficient is negative} \\ f(x, y) + \nabla^2 f(x, y) & \text{If centre coefficient is positive} \end{cases}$$

$$g(x, y) = 5f(x, y) - f(x+1, y) - f(x-1, y) - f(x, y+1) - f(x, y-1)$$

Laplacian Image Enhancement



- In the final sharpened image edges and fine detail are much more obvious

Also called edge crispening

0	-1	0
-1	5	-1
0	-1	0

Mask 1

-1	-1	-1
-1	9	-1
-1	-1	-1

Mask 2



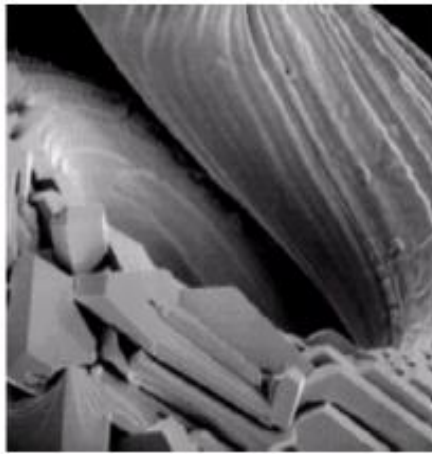
Original Image



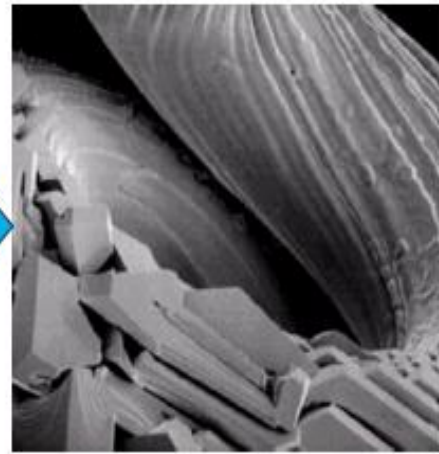
Mask 1



Mask 2



0	-1	0
-1	5	-1
0	-1	0



UNSHARP MASKING AND HIGH BOOST FILTERING

- Subtracting a blurred version of image from itself : Unsharp masking

$$f_s(x, y) = f(x, y) - \bar{f}(x, y)$$

- Generalization of UM: High boost filtering

$$\left. \begin{aligned} f_{hb}(x, y) &= Af(x, y) - \bar{f}(x, y) \\ f_{hb}(x, y) &= (A-1)f(x, y) + f(x, y) - \bar{f}(x, y) \\ f_{hb}(x, y) &= (A-1)f(x, y) + f_s(x, y) \end{aligned} \right\} \quad A \geq 1$$

- Using Laplacian

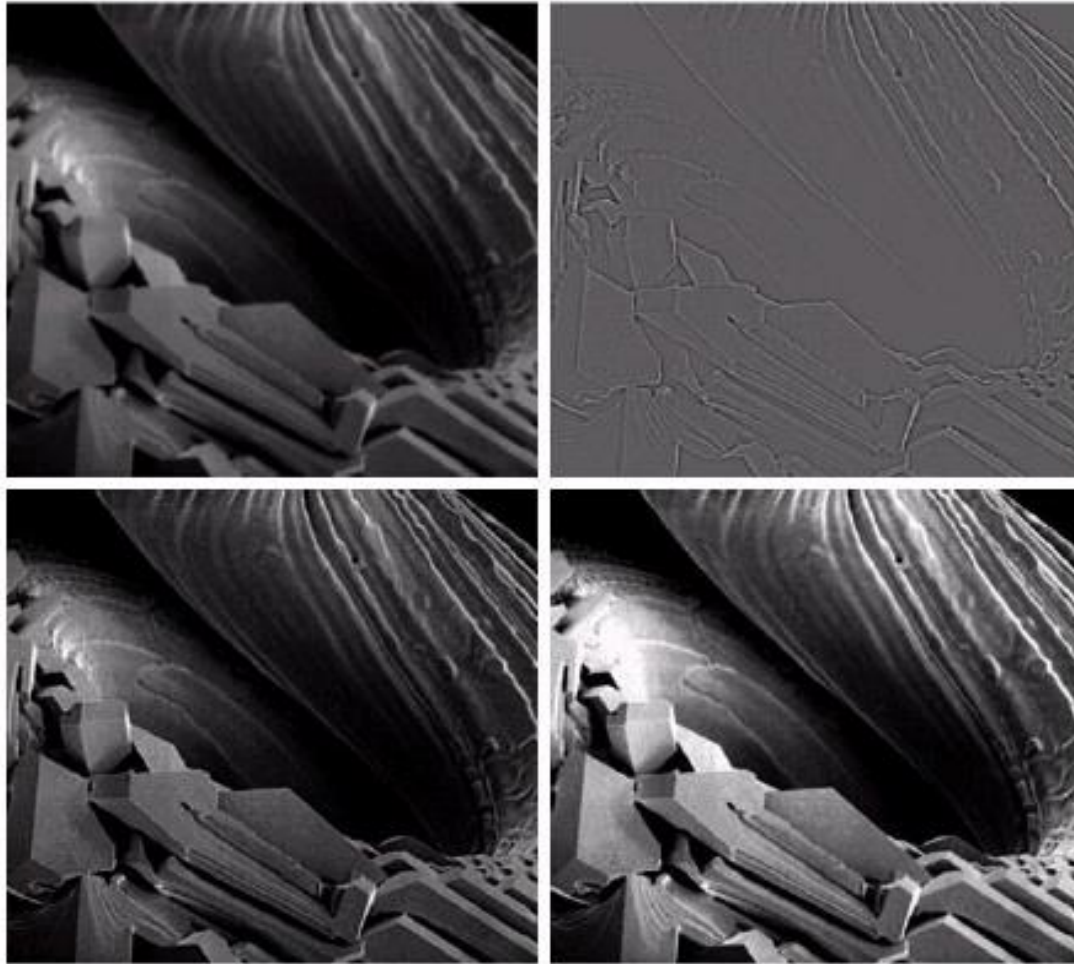
$$f_{hb}(x, y) = \begin{cases} Af(x, y) - \nabla^2 f(x, y) & \text{If centre coefficient is -ve} \\ Af(x, y) + \nabla^2 f(x, y) & \text{If centre coefficient is +ve} \end{cases}$$

0	-1	0	-1	-1	-1
-1	$A + 4$	-1	-1	$A + 8$	-1
0	-1	0	-1	-1	-1

a b
c d

FIGURE 3.43

(a) Same as Fig. 3.41(c), but darker.
(b) Laplacian of (a) computed with the mask in Fig. 3.42(b) using $A = 0$.
(c) Laplacian enhanced image using the mask in Fig. 3.42(b) with $A = 1$. (d) Same as (c), but using $A = 1.7$.



USING FIRST-ORDER DERIVATIVE FOR (NONLINEAR) IMAGE SHARPENING – THE GRADIENT

- First derivative of an image is implemented using the magnitude of the gradient
-
- For a function $f(x,y)$, the gradient of f at position (x,y) is defined as

$$\nabla \mathbf{f} = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

- The Magnitude of the Gradient

$$G[f(x, y)] = \text{mag}(\nabla \mathbf{f})$$

$$G[f(x, y)] = [G_x^2 + G_y^2]^{1/2}$$

$$G[f(x, y)] = \left[\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right]^{1/2}$$

$$G[f(x, y)] \approx |G_x| + |G_y|$$

The computational burden of implementing this equation over an entire image is not trivial.

It is common practice to approximate the magnitude of the gradient by using absolute values instead of squares and square roots.

The Direction Angle of the Gradient

- The direction angle of the gradient is given by

$$\alpha(x, y) = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

- The gradient angle is measured with respect to the positive direction of the x-axis.
- The values of G_x and G_y can be approximated by

$$G_x = (z_8 - z_5) \qquad G_y = (z_6 - z_5)$$

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

spatial mask

z	z	z	z	z	z	z	z
z	z_1	z_2	z_3	z	z	z	z
z	z_4	z_5	z_6	z	z	z	z
z	z_7	z_8	z_9	z	z	z	z
z	z	z	z	z	z	z	z
z	z	z	z	z	z	z	z
z	z	z	z	z	z	z	z

image

Roberts Cross-Gradient Operators

- One of the simplest ways to implement a first-order partial derivative

$$G_x = (z_9 - z_5) \quad G_y = (z_8 - z_6)$$

- The gradient can be computed as:

$$G[f(x, y)] = \left[(z_9 - z_5)^2 + (z_8 - z_6)^2 \right]^{1/2}$$

- The gradient can be approximated to:

$$G[f(x, y)] \approx |z_9 - z_5| + |z_8 - z_6|$$

Roberts Operator

-1	0	0	-1
0	1	1	0
G_x		G_y	

Masks of even size are awkward to implement because they do not have clear center.



Prewitt Operator

- An approach using masks of size 3x3 is given by:

$$G_x = (z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)$$

$$G_y = (z_7 + z_8 + z_9) - (z_1 + z_2 + z_3)$$

$$G[f(x, y)] = |(z_7 + z_8 + z_9) - (z_1 + z_2 + z_3)| + \\ |(z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)|$$

- The difference between the first and third rows of the 3 image region approximates the derivative in the x-direction, and the difference between the third and first columns approximates the derivative in the y-direction.

Prewitt Operator

-1	0	1	-1	-1	-1
-1	0	1	0	0	0
-1	0	1	1	1	1

G_x

G_y



Sobel Operator

- A slight variation of these two equations uses a weight of 2 in the center coefficient:

$$G_x = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

$$G_y = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

$$G[f(x, y)] = |(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)| + |(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)|$$

A weight value of 2 is used to achieve some smoothing by giving more importance to the center point

Sobel Operator

-1	0	1
-2	0	2
-1	0	1

G_x

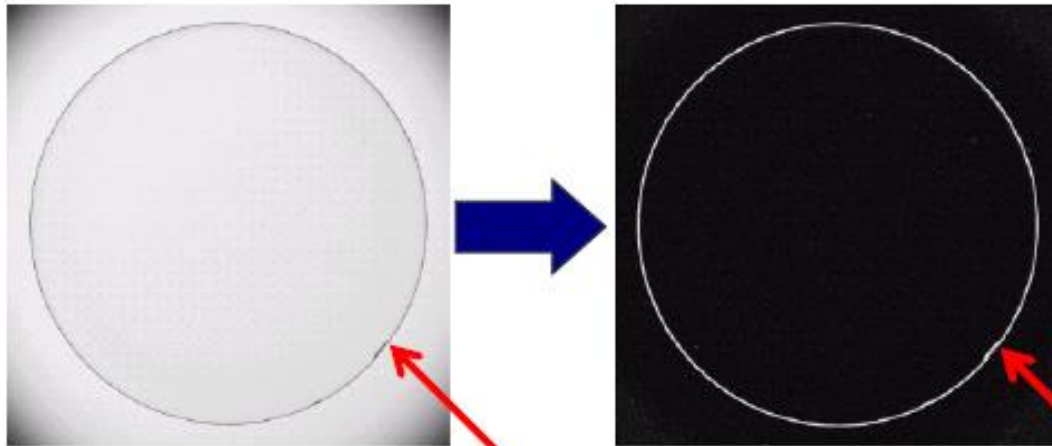
-1	-2	-1
0	0	0
1	2	1

G_y



Sobel Example

Sobel filters are typically used for edge detection



An image of a contact lens which is enhanced in order to make defects more obvious

Sobel Example

